

A LOW-POWER HIGH-SPEED DYNAMIC COMPARATOR WITH A TRANSCONDUCTANCE-ENHANCED LATCHING STAGE

MINTU KUMAR YADAV
department of Electrical Engineering
IIT Ropar
 Rupnagar, Punjab-140001
 2025EEM1026@iitrpr.ac.in

Abstract—The proposed work investigates a latch-driven dynamic comparator aimed at achieving low power consumption and fast switching performance under a 1.2 V supply condition. A conventional dynamic comparator is first implemented and evaluated using DC and transient simulations over a duration of 2 ns with a load capacitance of 20 fF. The baseline design shows a propagation delay of nearly 380 ps.

To enhance the performance, a modified comparator architecture incorporating a transconductance enhanced latching stage is developed and analyzed under the same operating conditions. The proposed design achieves a significantly lower delay of approximately 120 ps at the schematic level. The layout of the proposed circuit is realized within an area of $16\ \mu\text{m} \times 15\ \mu\text{m}$ and validated through DRC, LVS, and PEX verification steps. Post layout simulations indicate a delay of around 177 ps, demonstrating consistency with pre layout results.

Overall, the proposed comparator offers a notable reduction in delay while preserving low-power operation, making it a suitable candidate for high speed analog to digital (ADC) converter applications.

KEYWORDS

Dynamic comparator, high speed, low power, two stage comparator [8].

I. INTRODUCTION

High speed ADCs often employ dynamic comparators built with latch architectures. [1]–[3], memory sensing circuits [5], and digital interfaces due to their ability to operate with low static power and high speed and built-in self-test (BIST) applications [6]. Unlike static comparators, dynamic comparators rely on positive feedback and dynamic biasing, enabling faster operation and significantly low static power consumption [8].

In a conventional single-stage dynamic comparator, the input stage and latch are connected in a stacked manner. This structure increases the minimum voltage required for proper operation, thereby demanding larger voltage headroom.

II. CONVENTIONAL TWO-STAGE DYNAMIC COMPARATOR

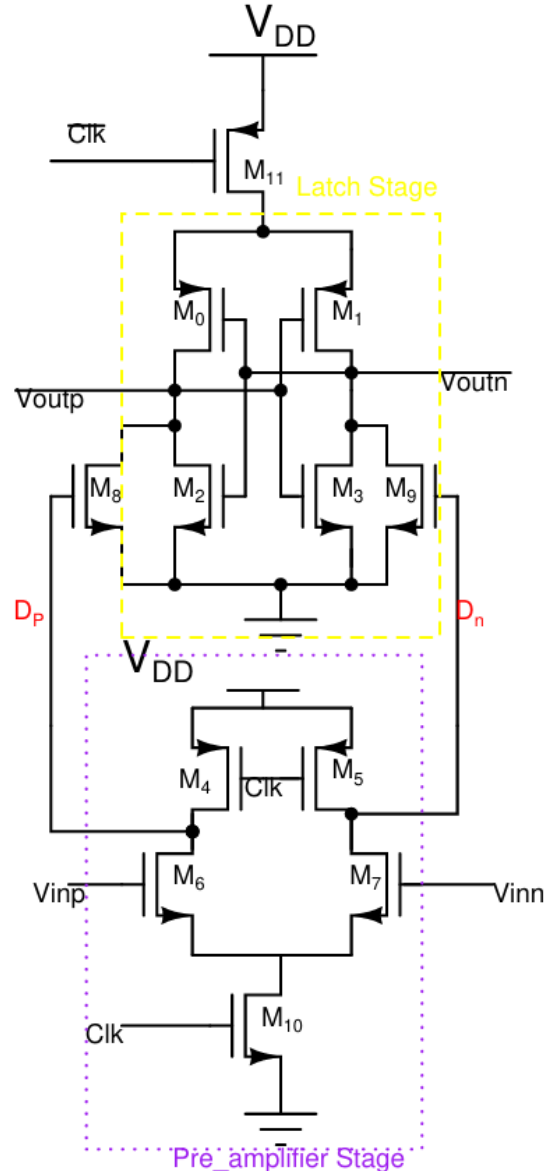


Fig. 1. Schematic of the conventional two stage dynamic comparator showing the preamplifier and latch stages.

Fig. 1 shows a dual stage dynamic comparator structure (circuit diagram). It is composed of two main parts: a preamplifier (Purple One) and a latch stage (Yellow One). In the preamplifier, a relatively small tail current is used to minimize the input referred offset voltage. Here, the latch is designed for higher current in order to achieve faster switching and improve the overall speed of the comparator.

TABLE I
TRANSISTOR DIMENSIONS OF THE CONVENTIONAL COMPARATOR

Transistor ID	Size (μm)
M0–M1	2 / 0.18
M2–M3	0.5 / 0.18
M4–M5	1.8 / 0.18
M6–M7	4 / 0.18
M8–M9	2 / 0.18
M10	0.44 / 0.18
M11	2 / 0.18

A. Reset Mode

When the circuit is in *reset phase*, clock should be low ($CLK = 0$), transistors M_4 and M_5 conduct, whereas M_{10} and M_{11} remain off. Under this condition, the internal nodes D_n and D_p are driven to the supply voltage V_{DD} [8]. At the same time, the output nodes V_{out_p} and V_{out_n} are pulled down to zero through M_8 and M_9 . This process initializes the circuit for the subsequent comparison phase.

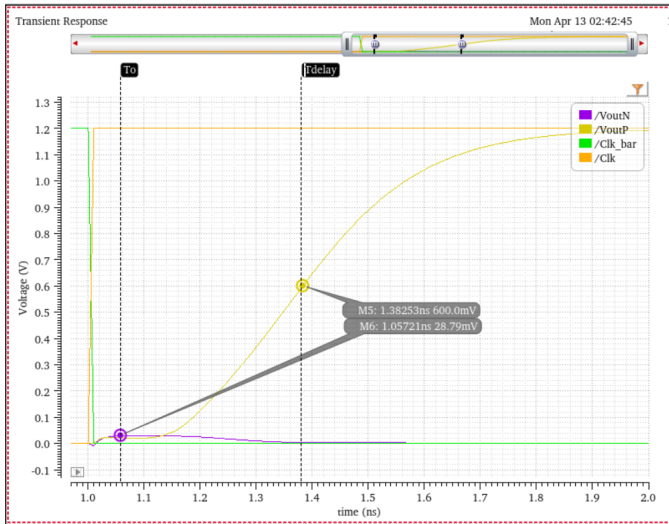


Fig. 2. It is Transient response of the conventional comparator illustrating output voltages during operation at 500 Mhz

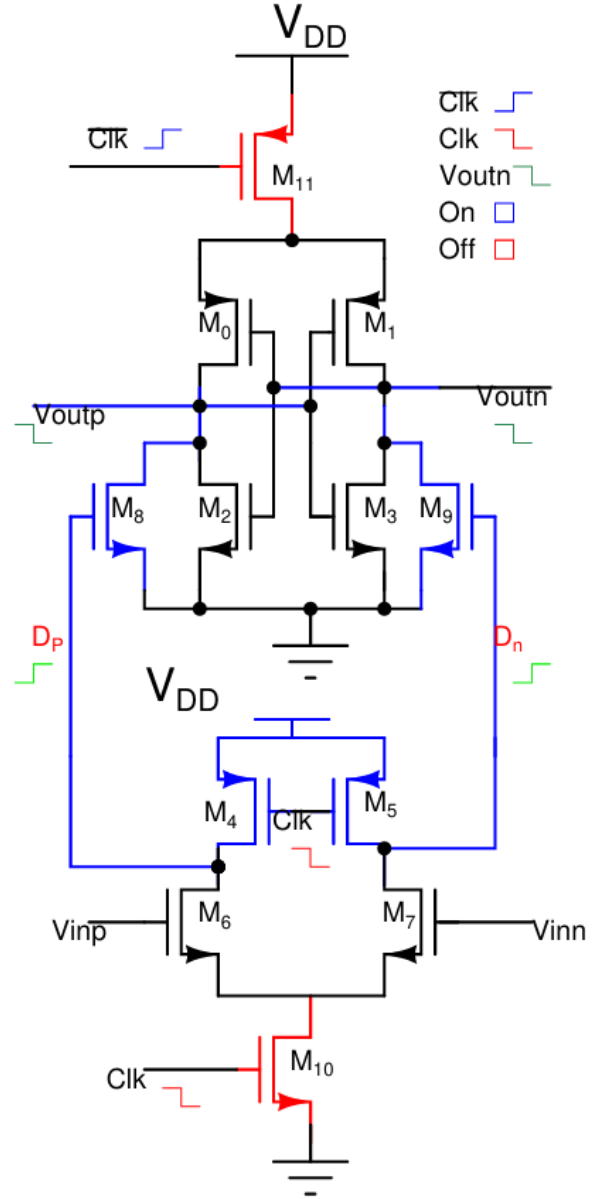


Fig. 3. It is circuit presentation when $Clk = 0$.

B. Comparison Mode

As the clock becomes high ($Clk = V_{DD}$), this configuration reaches to the *comparison mode*, it is showing in Fig. 2, in this phase, M_{10} and M_{11} are turning on, while M_4 and M_5 are turning off. The nodes potential D_n and D_p start to discharge with different currents depending on the input voltages V_{inn} and V_{inp} . This creates a small voltage difference between D_n and D_p [8].

This difference is transferred to the output nodes V_{out_p} and V_{out_n} and is quickly amplified by the cross-coupled latch due to positive feedback, producing a full logic-level output.

The two-stage structure offers profit such as reduced kickback noise, better operation at low supply, and improved speed and offset.

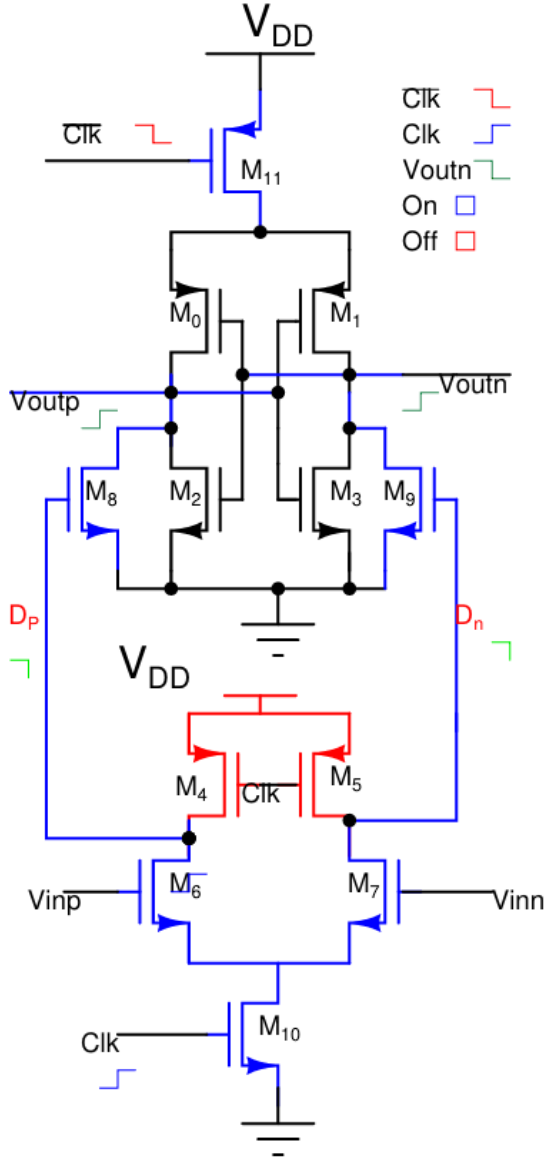


Fig. 4. Circuit behavior of the conventional comparator during evaluation($Clk = V_{DD}$)

C. Delay in Conventional Comparator

As we know delay time is the time took for output difference become $0.5V_{DD}$. This is divided in two parts:

- t_o : –

The time interval t_o is given by time span beginning with the rising transition of the clock to the instant when the output nodes V_{out_p} and V_{out_n} reach the threshold voltage V_{THN} . During this interval, the pMOS transistors M_0 and M_1 operate in the saturation region, whereas the nMOS transistors M_2 and M_3 remain in the cut off region [8].

Under these conditions, the effective transconductance $g_{m,eff}$ of the latch is primarily determined by M_0 and M_1 . Accordingly, the time t_o can be expressed as

$$t_o = \frac{V_{THN}C_{OUT}}{I_p} \approx \frac{V_{THN}C_{OUT}}{2\mu_p C_{ox} \frac{W_p}{L_p}} \approx \frac{L_p}{g_{m,eff}^2}$$

The overall transconductance is obtained by summing the contributions of PMOS and NMOS devices:

$$g_{m,eff} = g_{mp} + g_{mn}$$

where:

- C_{ox} denotes the gate oxide capacitance
- μ_p represents the hole mobility
- V_{THN} : Represents Threshold voltage of M_2 and M_3
- C_{OUT} : Load capacitances connected to output node
- I_p : Current of transistor M_1
- g_{mp} : Transconductance of pMOS transistor M_1
- g_{mn} : Transconductance of nMOS transistor
- W_p, L_p : Width and length of M_1

Since M_3 is off during this phase, g_{mn} is negligible, and hence $g_{m,eff} \approx g_{mp}$.

- t_{latch} : – It corresponds to the time interval from the end of t_o until the output voltage reaches $(V_{DD}/2)$. It is Delay for Latching Stage which is given below

$$t_{latch} = \frac{C_o}{g_{m,eff}} \ln \left(\frac{V_{out}}{V_0} \right) = \frac{C_{out}}{g_{m,eff}} \ln \left(\frac{V_{dd}/2}{V_0} \right)$$

Where

- V_0 : The voltage difference across the output nodes at $t = t_o$, corresponding to the condition when the nMOS device is non-conducting

During the regeneration phase, all transistors in the cross-coupled structure are turned on so now all transistor is in saturation so The effective transconductance $g_{m,eff}$ associated with the comparator is therefore expressed as:

$$g_{m,eff} = g_{mp} + g_{mn}$$

$$g_{m,eff} = \mu_p C_{ox} \frac{W_p}{L_p} (|V_{gsP}| - V_{ThP}) + \mu_n C_{ox} \frac{W_n}{L_n} (V_{gsN} - V_{ThN})$$

$$t_{delay} = \frac{V_{THN}C_{OUT}}{2\mu_p C_{ox} \frac{W_p}{L_p}} + \frac{C_{OUT}}{g_{m,eff}} \ln \left(\frac{V_{dd}/2}{V_0} \right)$$

From the above analysis, it is evident that increasing the effective transconductance $g_{m,eff}$ is a main way to reduce the overall delays of the comparator, mainly during the initial phase t_o [8]. A higher $g_{m,eff}$ enhances the regeneration speed, thereby shortening the decision time.

However, in traditional two-stage dynamic comparators, only the pMOS devices contribute to $g_{m,eff}$ at the beginning of the comparison phase, as the nMOS transistors remain off (inactive). This limitation results in slower initial regeneration and prolongs the metastable condition of the latch.

Moreover, increasing $g_{m,eff}$ improves speed, it approximately requires larger transistor sizes or higher bias currents, which in turn increases power consumption and area. Therefore, a trade-off exists between delay reduction and power efficiency, making it essential to optimize the transconductance without significantly degrading energy performance.

III. PROPOSED DYNAMIC COMPARATOR

A. CIRCUIT DESCRIPTION

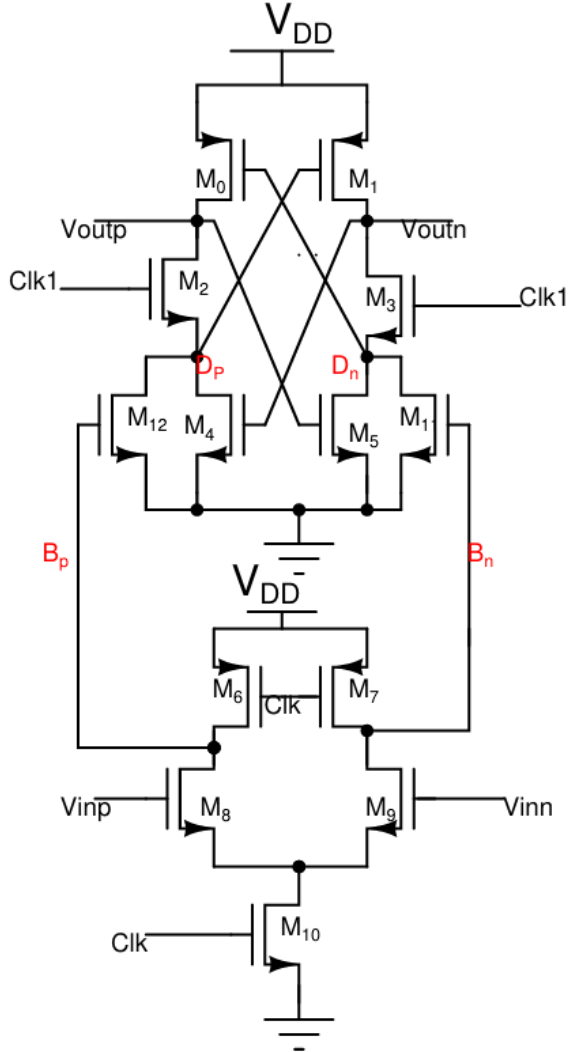


Fig. 5. Schematic representation of the developed dynamic comparator featuring a transconductance enhanced latching mechanism.

Reset Mode

In Fig.5, Throughout *reset mode* ($Clk = 0$), the node B_n and B_p are precharged to the supply voltage V_{DD} . In this state, transistors M_{12} , M_{11} , M_0 , and M_1 are turned On, while M_2 and M_3 remain Off. The nodes D_n and D_p are pull down to ground by M_6 and M_7 [8]. At the same time, the output port V_{out_p} and V_{out_n} are charged to V_{DD} using M_0 and M_1 , thereby initializing the circuit for

the subsequent phase.

In the *reset phase*, the proposed regenerative structure offers a significant improvement over the traditional cross coupled latch. In conventional circuit, only two transistors are biased in the strong inversion region, while the rest of two are remaining in cutoff, thereby limiting their contribution to the overall transconductance and slowing down at the beginning. In contrast, the proposed structure ensures that all cross coupled transistors M_0 , M_1 , M_4 , and M_5 operate in the strong inversion region during reset mode. This enables all devices to contribute to the effective transconductance, enhancing the $g_{m,eff}$ of the latch for regeneration. As a result, a faster response and reduced metastability can be achieved during the subsequent comparison phase.

Comparison Mode

In Fig.6, At the start of the *comparison phase*, the developed regenerative stage has a larger effective transconductance than the traditional design. This is because more transistors are active and take part in the regeneration process. Due to this, the output voltage changes faster and the comparator makes a quicker decision.

Also, the faster operation reduces the time for which the circuit stays in the metastable region. This helps in lowering the energy consumption, since the circuit spends less time in unstable states.

In addition, the clock signal $Clk1$ is slightly delayed compared to Clk . Because of this delay, transistors M_2 and M_3 turn on a little later. This reduces the short circuit current via M_6 and M_7 at the positive edge of Clk , which improves the power efficiency of the circuit [8].

The Final delay of the proposed dynamic comparator can be

$$t_{delay} = t_{opm} + t_0 + t_{latch}$$

During the time interval t_0 , the output nodes V_{out_p} and V_{out_n} start pulling down from V_{DD} , while the internal nodes D_p and D_n begin to rise from ground. At the beginning of comparison mode, all transistors in the cross coupled structure (M_0 , M_1 , M_4 , M_5) are turned on and operate in the linear region.

By this condition, the effective transconductance of the latch depends on both nMOS and pMOS transistors and can be expressed as

$$g_{m,eff} = g_{mn} + g_{mp} = \mu_n C_{ox} \frac{W_n}{L_n} V_{dsn} + \mu_p C_{ox} \frac{W_p}{L_p} |V_{dsp}|$$

When the delayed clock signal ($Clk1$) becomes high, transistors M_2 and M_3 will be turned on. This quickly pulls V_{out_p} and V_{out_n} toward D_p and D_n , which increases V_{dsn} and $|V_{dsp}|$ and improves the regeneration speed.

In contrast, in a traditional positive feedback latch circuit, only the pMOS transistors are active and operate in the saturation region. For this case, the effective transconductance is given by

$$g_{m,eff} = g_{mp} = \mu_p C_{ox} \frac{W_p}{L_p} (|V_{gsP}| - V_{ThP})$$

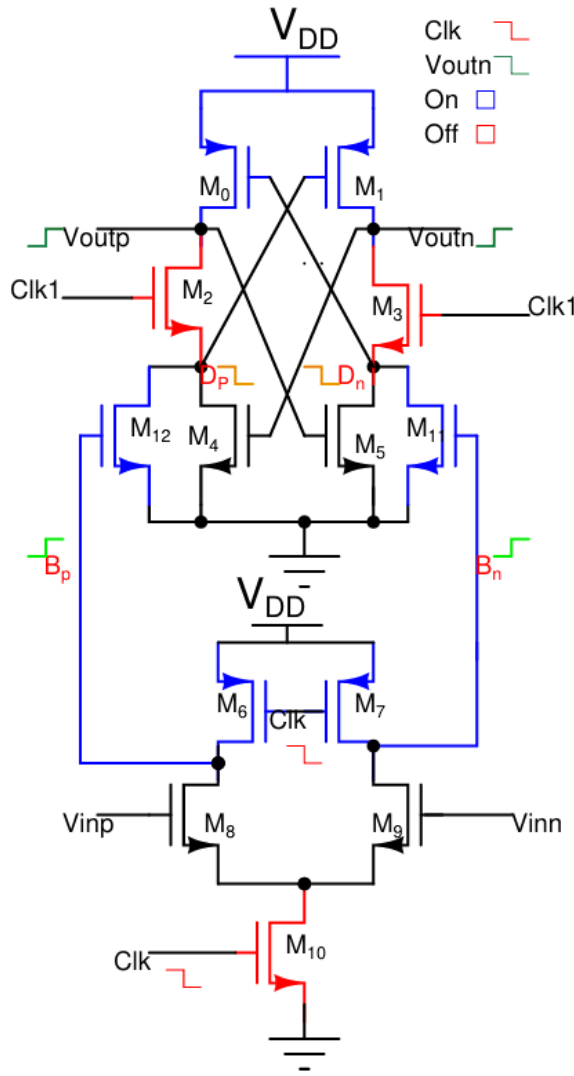


Fig. 6. Reset phase operation of the proposed dynamic comparator showing initial node conditions.

Comparison Mode

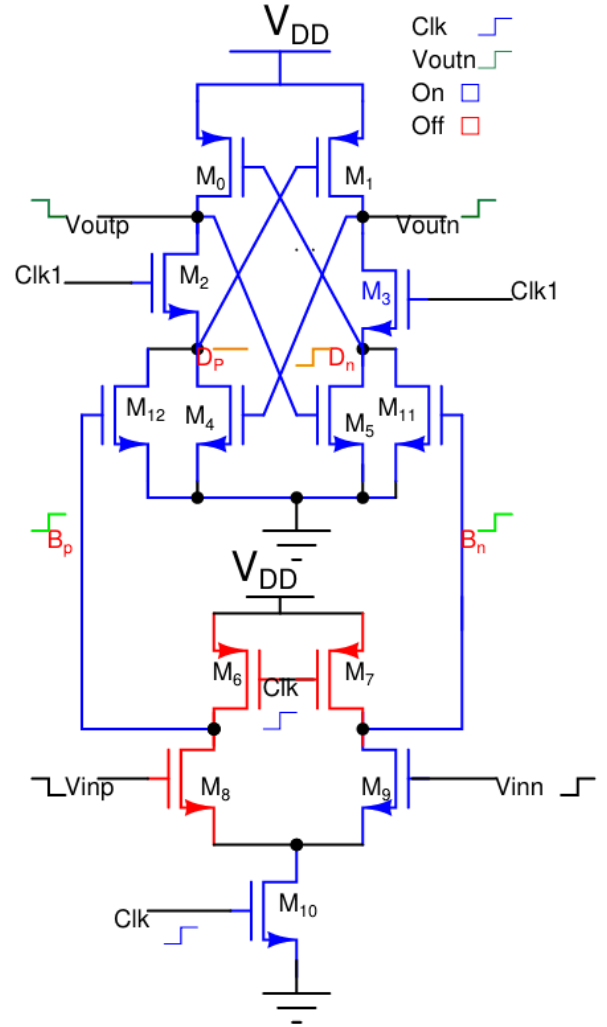


Fig. 8. Comparison phase operation of the proposed dynamic comparator illustrating enhanced regeneration.

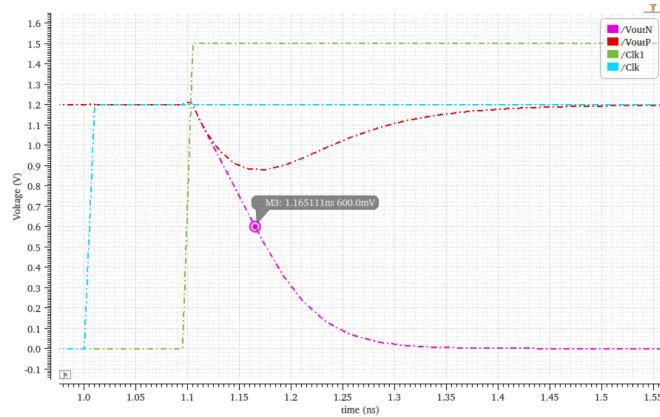


Fig. 7. Transient response of the dynamic comparator showing output voltages (VoutP, VoutN) and clock signals (Clk, Clk1), highlighting the switching behavior and propagation delay (165ps) at 500 MHz.

IV. DESIGN CONSIDERATIONS

For removing the issue which was in convention comparator we have to use two more switches. The key feature of the design is the modified cross coupled latch along with the switches M_2 and M_3 in Propose Comparator which can be seen in fig.4.

During the comparison mode, transistors M_2 and M_3 are turned on by delay clock ($Clk1$) and help in discharging the output nodes, which depends on V_{inn} and V_{inp} . For proper operation, the drain to source voltage of these switches should be kept as small as possible to allow to completely discharge of the output port.

If this voltage is not decreasing, then the current flow through the switches may be limited, which can help to complete discharge of the output nodes. This can slow down the regeneration process and reduce the performance of the comparator. Therefore, proper sizing and design

of M_2 and M_3 are necessary to achieve fast and reliable operation.

TABLE II
TRANSISTOR DIMENSIONS OF THE CONVENTIONAL COMPARATOR

Transistor ID	Size (μm)
M0-M1	2 / 0.18
M2-M3	1 / 0.18
M4-M5	0.5 / 0.18
M6-M7	1.8 / 0.18
M8-M9	4 / 0.18
M10	0.44 / 0.18
M11-M12	2 / 0.18

V. EXPERIMENTAL RESULTS

In order to evaluate how the developed comparator performs, it is compared with the traditional latch comparator through simulation (Transient simulation). Both circuits are designed in $0.18\mu m$ CMOS process (Scldpk) using a supply voltage of $1.2V$ and an load capacitance of $20fF$ [8].

For a more accuracy, the transistor sizes of both designs are kept similar so that both circuits operate under same conditions. This ensures that the comparison is done at circuit design level only.

Furthermore, the clock frequency is set to 500 MHz to analyze the high-speed performance of the comparators. Under these conditions, any variation in results is mainly due to the proposed design.

Layout

I have design layout of the propose comparator which govern by standard VLSI design rules. After completing the layout, it has passed by several verification steps to ensure the correctness of the design.

First, Design Rule Check (DRC) is carried out to verify that the layout satisfies all the fabrication rule of the technology. The results show no violations, and indicating that the layout is clean.

Next, Layout Versus Schematic (LVS) verification is performed to check layout align perfectly to the schematic design. The LVS results are also clean, it has confirmed that layout is corresponding to device connections and structure. After that, Parasitic Extraction (PEX) is performed to include the effect of parasitic resistances and capacitances present in the layout, which helps in obtaining more accurate simulation results.

All these steps are completed using the Calibre tool with the SCL 0.18 μm PDK, and the results are found to be clean. This confirms that the layout is correct and ready for further analysis or fabrication.

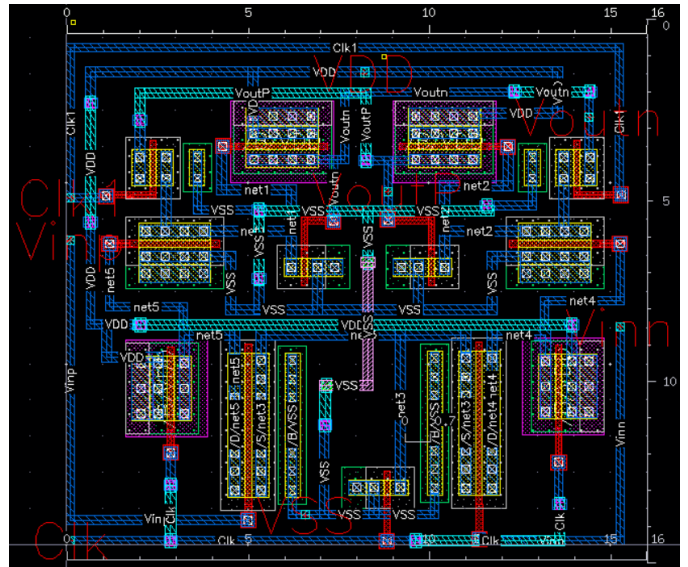


Fig. 9. A 0.18 μ m CMOS process with the SCL PDK has been used to implement the layout of the develop dynamic comparator

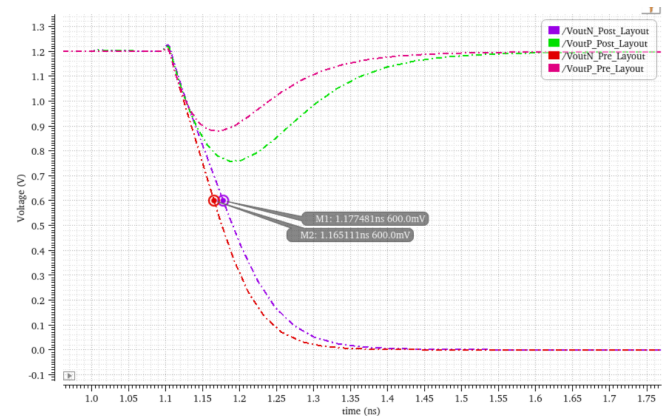


Fig. 10. Before and after layout extraction, transient responses of the developed comparator demonstrating performance consistency at 500 Mhz.

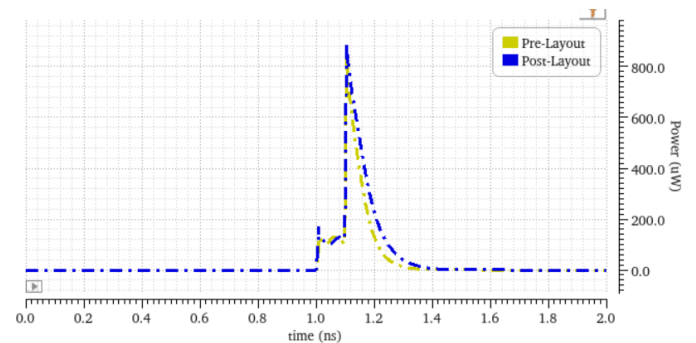


Fig. 11. Power plot with time for the developed comparator, showing schematic and physical level behavior

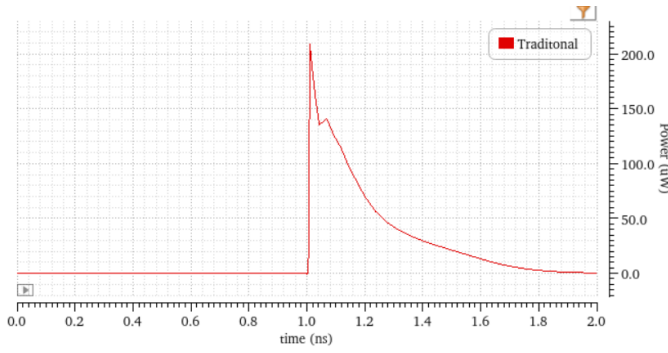


Fig. 12. Power plot with time for traditional comparator

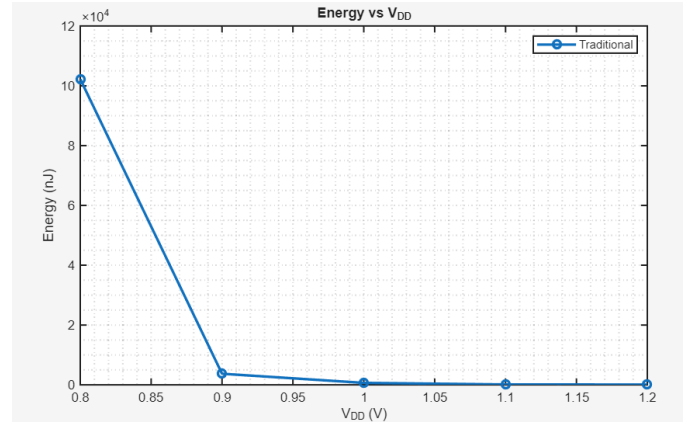


Fig. 15. Energy vs VDD for traditional comparator

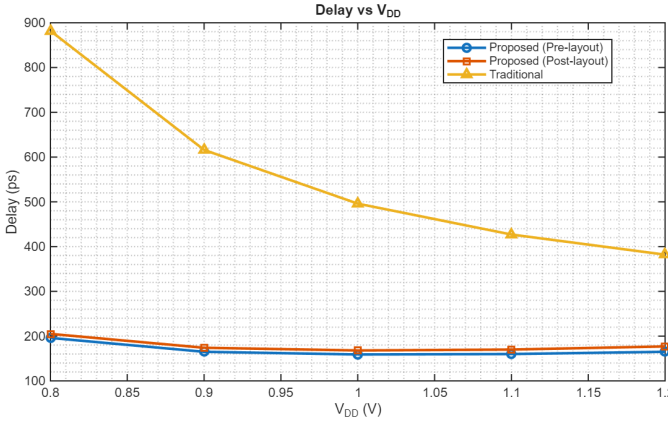


Fig. 13. Effect of supply voltage on delay for the developed comparator under pre-layout and post-layout conditions, compared with a traditional approach.

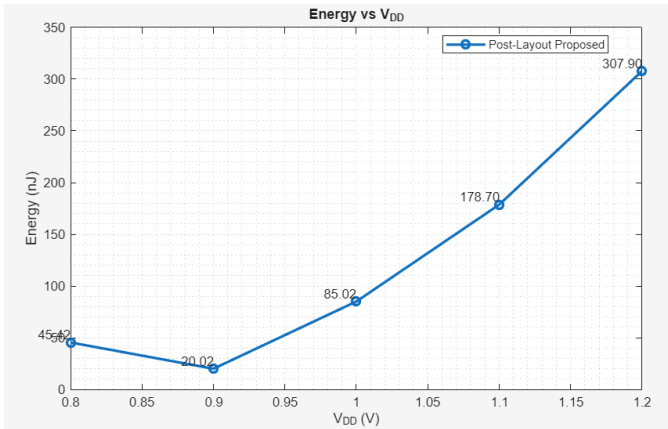


Fig. 14. Energy vs VDD for developed comparator

VI. CONCLUSION

In this work, I have designed the proposed dynamic comparator with the aim of improving the effective transconductance of the latch. The circuit uses a modified cross coupled structure, which helps in achieving better performance. From both schematic and physical level simulations, it is observed that the developed design shows faster operation compared to the traditional comparator. The delay is reduced due to faster regeneration, and the energy consumption is also less than conventional one since the circuit spends less time in the metastable region.

The layout is successfully completed and verified using DRC, LVS, and PEX, ensuring that the design is correct and reliable. The physical level simulation results also follow the expected trend and are close to the reference design.

Overall, the developed comparator provides an effective trade off among speed, power, and area. It is suitable for applications requiring both low voltage and high speed, and its simple structure makes it practical for implementation.

REFERENCES

- [1] L. B. Leene and T. G. Constantinou, "A 0.016 mm² 12 b SAR with 14 fJ/conv. for ultra low power biosensor arrays," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 64, no. 10, pp. 2655–2665, Oct. 2017.
- [2] K. Yoshioka, T. Sugimoto, N. Waki, S. Kim, D. Kurose, H. Ishii, M. Furuta, A. Sai, and T. Itakura, "28.7 A 0.7V 12b 160 MS/s 12.8 fJ/conv-step pipelined-SAR ADC in 28nm CMOS with digital amplifier technique," in *IEEE ISSCC Dig. Tech. Papers*, Feb. 2017, pp. 478–479.
- [3] C. Wulff and T. Ytterdal, "A compiled 9-bit 20-MS/s 3.5-fJ/conv-step SAR ADC in 28-nm FDSOI for Bluetooth low energy receivers," *IEEE J. Solid-State Circuits*, vol. 52, no. 7, pp. 1915–1926, Jul. 2017.
- [4] K. Huang, D. Luo, Z. Wang, X. Zheng, F. Li, C. Zhang, and Z. Wang, "A 190 mW 40 Gbps SerDes transmitter and receiver chipset in 65 nm CMOS technology," in *Proc. IEEE Custom Integr. Circuits Conf. (CICC)*, Sep. 2015, pp. 1–4.
- [5] D. Schinkel, E. Mensink, E. Klumperink, E. van Tuijl, and B. Nauta, "A double-tail latch-type voltage sense amplifier with 18ps setup+hold time," in *IEEE Int. Solid-State Circuits Conf. Dig. Tech. Papers*, Feb. 2007, pp. 314–605.

- [6] S. Lee, C. Shi, J. Wang, A. Sanabria, H. Osman, J. Hu, E. Sánchez-Sinencio, "A built-in self-test and in situ analog circuit optimization platform," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 65, no. 10, pp. 3445–3458, Oct. 2018.
- [7] B. Razavi, "Charge steering: A low-power design paradigm," in *Proc. Custom Integr. Circuits Conf. (CICC)*, Sep. 2013, pp. 1–8.
- [8] Y. Wang, M. Yao, B. Guo, Z. Wu, W. Fan and J. J. Liou, "A Low-Power High-Speed Dynamic Comparator With a Transconductance-Enhanced Latching Stage," in *IEEE Access*, vol. 7, pp. 93396–93403, 2019, doi: 10.1109/ACCESS.2019.2927514.
 keywords: Browsers;Crawlers;Navigation;Web mining;Web services;Servers;Dynamic comparator;high-speed;low-power;two-stage comparator,